Project

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2026-04-06

Background

Problem Formulation

The NBA Draft is an annual event where all 30 NBA teams select eligible players to join their rosters, serving as the primary pathway for new talent to enter the league. The draft consists of two rounds, with each team holding one pick per round, for a total of 60 selections. The order in which teams pick is not arbitrary: teams that performed worse in the previous season are rewarded with earlier picks, giving struggling franchises the opportunity to rebuild through high-potential young talent. Specifically, the 14 lowest-ranked teams enter a weighted lottery to determine the first 14 picks of the first round, with worse records receiving higher odds. The remaining picks in both rounds follow reverse standings order — the worst team picks first, the best team picks last. Prior to the draft, teams evaluate prospects through the NBA Draft Combine, private workouts, and scouting reports. Scouts assess a wide range of factors including physical measurements, athletic testing, skill level, and projected development trajectory. The result is an internal team ranking that ultimately determines which player a team selects when their pick arrives.

Pick position carries substantial consequences for a player's career. First-round selections receive fully guaranteed multi-year contracts, whereas second-round picks are not guaranteed and must compete for roster spots with little job security. This distinction makes the first-round versus second-round boundary a meaningful and practically important threshold to study.

Given this context, understanding what physical attributes are associated with first-round selection is of both scientific and practical interest. Height, in particular, has long been regarded as a fundamental advantage in basketball — taller players are better positioned to score near the basket, contest shots, and control the boards. It is therefore natural to ask whether teams systematically favour taller players when making first-round selections. Our scientific question is:

> **Does player height have a positive effect on the probability of being selected in the first round of the NBA Draft?**

Mathematically, we model this via Bayesian logistic regression, where the binary outcome (first round = 1, second round = 0) is regressed on player height. The key quantity of interest is the **posterior probability that the slope on height is positive**:

$$P(\beta_{\text{height}} > 0 \mid \mathbf{Y})$$

Team Contributions

Public repository link: https://github.com/ymyms/stat-405-project

Literature Review

...

Data Cleaning

```
nba <- read.csv("all_seasons.csv")
```

The raw dataset contains 12,844 player-season observations across 22 variables, spanning the 1996–97 to 2022–23 NBA seasons. Each row represents one player in one season, meaning the same player may appear multiple times across different years.

```
recent_20 <- nba[nba$draft_year>=2006 & !(nba$draft_round== "Undrafted"),]
rookie_data <- recent_20 |>
  dplyr::arrange(player_name, season) |>
  dplyr::distinct(player_name, .keep_all = TRUE)
rookie_data$draft_round <- ifelse(rookie_data$draft_round==1,1,0)
```

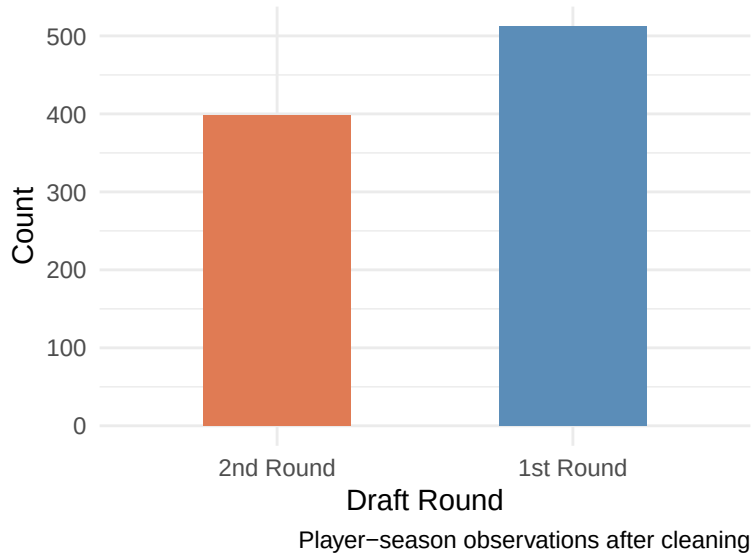
Four cleaning steps were applied to prepare the data for modelling. First, we restricted the data to players drafted in 2006 or later, retaining only the past decade of draft classes. This ensures that our findings reflect the preferences and evaluation criteria of the modern NBA. Second, players recorded as undrafted were excluded, since our research question concerns the distinction between first- and second-round picks specifically, undrafted players represent a qualitatively different pathway into the league. Third, to avoid the statistical issue of repeated observations per player (the same player's height and draft round appearing across multiple seasons), we retained only each player's earliest season. This gives one observation per unique player, ensuring the independence assumption of logistic regression is satisfied. Fourth, the `draft_round` variable was recoded as a binary indicator: 1 for first-round selections and 0 for second-round selections.

EDA

The goal of this section is to examine the distribution of player height and its relationship with draft round.

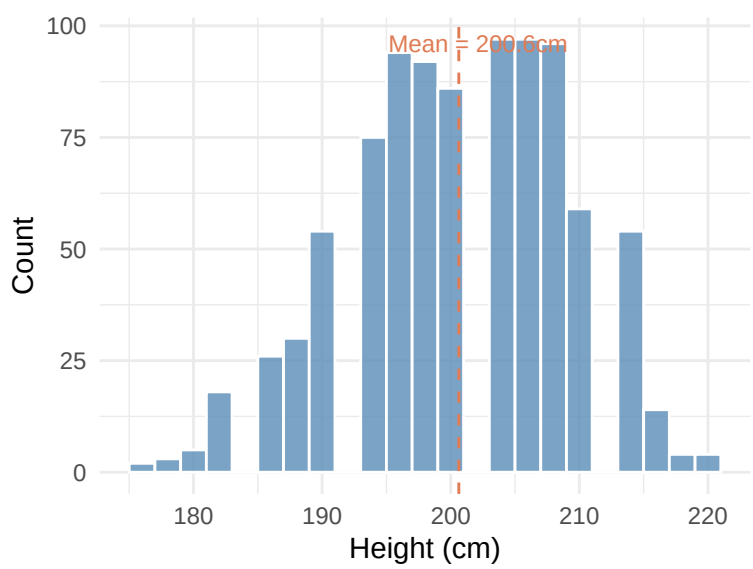
Distribution of Draft Round

```
ggplot(rookie_data,
       aes(x = factor(draft_round, labels = c("2nd Round", "1st Round")))) + geom_bar(fill = c("#E07B54
  labs(x = "Draft Round", y = "Count",
       caption = "Player-season observations after cleaning")+
  theme_minimal()
```



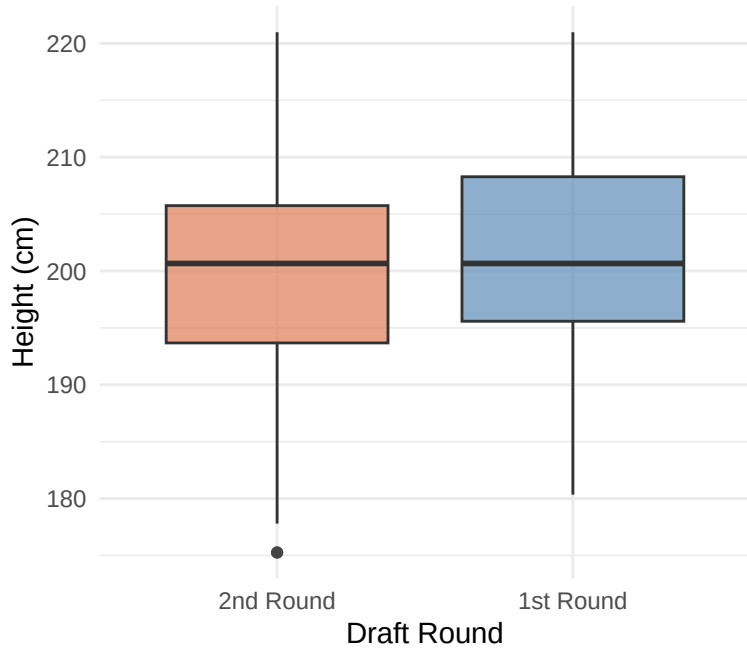
Distribution of Player Height

```
ggplot(rookie_data, aes(x = player_height)) +
  geom_histogram(binwidth = 2, fill = "#5B8DB8", color = "white", alpha = 0.8) +
  geom_vline(xintercept = mean(rookie_data$player_height),
             linetype = "dashed", color = "#E07B54") +
  annotate("text", x = mean(rookie_data$player_height) + 1.5,
           y = Inf, vjust = 2,
           label = paste0("Mean = ",
             round(mean(rookie_data$player_height), 1), "cm"),
           color = "#E07B54", size = 3) +
  labs(x = "Height (cm)", y = "Count") + theme_minimal()
```



Height by Draft Round

```
ggplot(rookie_data,
       aes(x = factor(draft_round, labels = c("2nd Round", "1st Round")),
           y = player_height,
           fill = factor(draft_round))) +
  geom_boxplot(alpha = 0.7, show.legend = FALSE) +
  scale_fill_manual(values = c("#E07B54", "#5B8DB8")) +
  labs(x = "Draft Round", y = "Height (cm)") + theme_minimal()
```




```
rookie_data %>%
  group_by(draft_round) %>%
  summarise(
    n        = dplyr::n(),
    mean_ht  = round(mean(player_height), 2),
    sd_ht    = round(sd(player_height), 2),
    med_ht   = round(median(player_height), 2)
  ) %>%
  mutate(draft_round = ifelse(draft_round == 1,
                              "1st Round", "2nd Round")) %>%
  rename(`Draft Round` = draft_round, N = n,
         Mean = mean_ht, SD = sd_ht, Median = med_ht)
```

```
## # A tibble: 2 x 5
##   `Draft Round`     N  Mean    SD Median
##   <chr>         <int> <dbl> <dbl>  <dbl>
## 1 2nd Round       398  200.  8.65   201.
## 2 1st Round       512  201.  8.16   201.
```

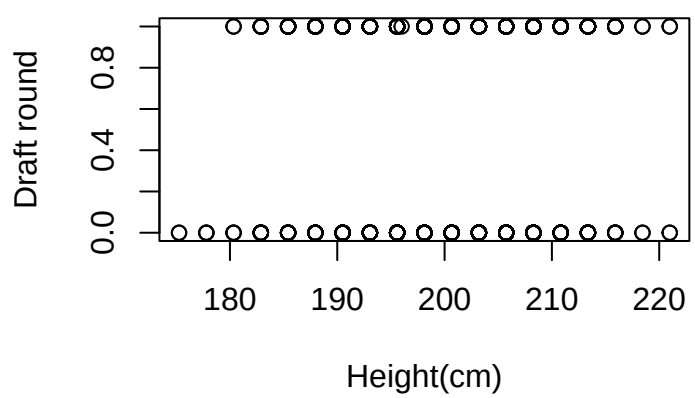
First-round picks are on average taller than second-round picks, though the difference is modest and the distributions overlap considerably. The EDA is consistent with a positive but uncertain slope in the logistic regression model.

```
glm(formula = draft_round~player_height,
       data = rookie_data, family = binomial)%>%  tidy(exponentiate = F)
```

```
## # A tibble: 2 x 5
##   term          estimate std.error statistic p.value
##   <chr>            <dbl>     <dbl>     <dbl>   <dbl>
## 1 (Intercept)    -3.32     1.61        -2.07  0.0388
## 2 player_height   0.0178   0.00800      2.22  0.0261
```

In the frequentist interpretation, since the p value for slope is smaller than 0.05 (0.029), and the slope is larger than 0 (0.027), we could conclude that there is a statistically significant positive association between the player's height and draft round. We could like to also verify this idea using the Bayesian approach.

```
plot(y=rookie_data$draft_round,x=rookie_data$player_height, ylab="Draft round",xlab="Height(cm)")
```



From the scatter plot, we can see that when player's height is below 185cm, it is very unlikely for the player to be drafted in the first round. Howeverm after 185cm, the nuisance becomes less obvious.

The model has distribution as following.

$$\text{slope} \sim N(0,10)\text{intercept} \sim N(0,10)\theta(\text{height}) = \text{logistic}(\text{intercept} + \text{height} \cdot \text{slope})Y \sim \text{Bernoulli}(\theta(\text{height}))$$

Model Fitting

```
data_list <- list(
  N = nrow(rookie_data),
  height = rookie_data$player_height,
```

```r
  y = rookie_data$draft_round
)
mod <- cmdstan_model("logistic_slope.stan")
fit <- mod$sample(
  data = data_list,
  seed = 1,
  chains = 2,
  iter_sampling = 2000,
  refresh = 0
)
```

```
## Running MCMC with 2 sequential chains...
##
## Chain 1 finished in 3.5 seconds.
## Chain 2 finished in 4.0 seconds.
##
## Both chains finished successfully.
## Mean chain execution time: 3.8 seconds.
## Total execution time: 7.8 seconds.
```

```r
fit$summary(variables = c("intercept", "slope","slope_positive"), "mean","sd", "rhat","ess_bulk", "ess_
```

```
## # A tibble: 3 x 6
##   variable           mean      sd  rhat ess_bulk ess_tail
##   <chr>             <dbl>   <dbl> <dbl>    <dbl>    <dbl>
## 1 intercept      -0.878   0.857    1.01     505.     550.
## 2 slope           0.00565 0.00428  1.01     504.     560.
## 3 slope_positive  0.906   0.292    1.00     565.      NA
```

```r
cat("Exponentiated value of the slope coefficient is", exp(mean(fit$draws("slope"))),".")
```

```
## Exponentiated value of the slope coefficient is 1.005667 .
```

The effective sample size is around 500 for both variables.

We choose to use 2 chains so we could check whether the chain mixes well later. From the table, we can conclude that the mean of slope is 0.03, and the mean of `slope_positive`, an indicator variable is 0.98.

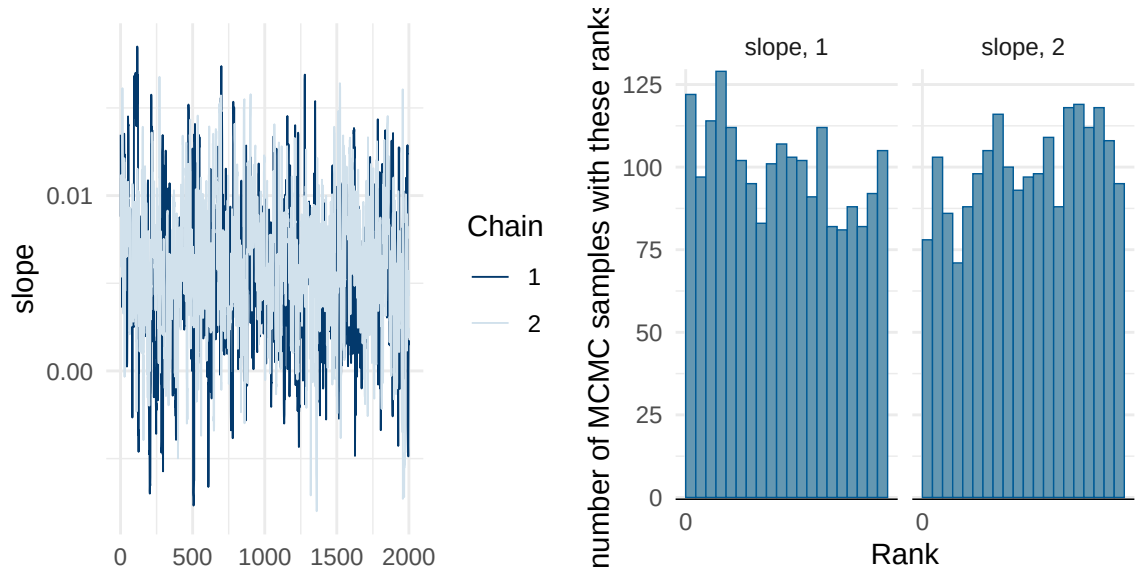
Since $E[1[slope > 0]|Y] = P(slope > 0|Y)$, this model effectively answered our question, as the probability of interest is substancially larger than 0.5. Thus, we conclude that height has a positive relationship with the chance of being drafted in the first round. Thus, being taller in height means higher probability of drafted earlier.

From the table, we can also conclude that a 1 cm increase in height is associated with a 2.6% increase in the odds of a player being drafted in the first round.

We would like to assess the goodness of fit using the following approach.

MCMC Diagnostics

```
trace <- mcmc_trace(fit$draws("slope")) + theme_minimal()
hist <- mcmc_rank_hist(fit$draws("slope")) +
  ylab("number of MCMC samples with these ranks") +
  theme_minimal()+
  scale_x_continuous(
  breaks = c(0, 5000, 10000, 15000, 20000),
  labels = c("0", "", "10000", "", "20000"))
gridExtra::grid.arrange(trace,hist,ncol=2)
```

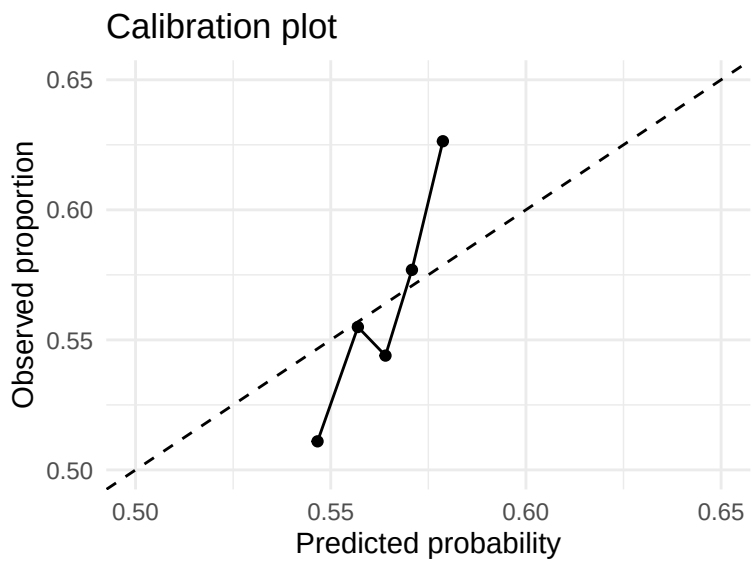


Here, we are using `slope` instead of `slope_positive` to check whether the chain mixes well. this is because that `slope` is a continuous variable , but `slope_positive` is binary. Thus, fitting the trace plots and histograms on this binary variable will most likely generate some vertical lines, which becomes harder to interpret. Since both variables are form the same model, if the plots for `slope` suggests fast mixing, then it means that the model for `slope_positive` mixes fast. From the trace plot, we can see that they two chains overlap a lot; from the histogram, we can see two approximately uniform histograms. Both plots suggests that our model is fast mixing.

Goodness of Fit

```
draws_df <- as_draws_df(fit$draws(c("intercept", "slope")))
rookie_data$p_hat <- sapply(
  1:nrow(rookie_data),
  function(i) {
    mean(plogis(draws_df$intercept + draws_df$slope * rookie_data$player_height[i]))})
rookie_data$bin <- ntile(rookie_data$p_hat, 5)
calibration_df <- rookie_data %>%
  group_by(bin) %>%
  summarise( mean_pred = mean(p_hat),
    obs_rate = mean(draft_round))
ggplot(calibration_df, aes(x = mean_pred, y = obs_rate)) +
  geom_point() + geom_line() +
  geom_abline(slope = 1, intercept = 0, linetype = "dashed") +
```

```r
  coord_cartesian(xlim = c(0.5, 0.65), ylim = c(0.5, 0.65)) +
  labs(x = "Predicted probability", y = "Observed proportion",
    title = "Calibration plot") + theme_minimal()
```



We can see that the majority of the probability center around 0.55 to 0.6, which suggests that although there is a positive relationship between height and draft round, the association is relatively weak.

Now, we would simulate the data and sere if the nominal coverage rate matches with the actual coverage rate. From, EDA, we see that the histogram of player height is approximately normal, centered roughly around 200cm, with a standard deviation around 10cm.

```r
set.seed(405)
simulate_data <- function(N,intercept, slope){
  height <- rnorm(N,200,10)
  y <- rbinom(N, 1, plogis(intercept+slope*height))
  data <- list(N=N,height=height,y=y)
  fit <- mod$sample(
  data = data, chains = 2, iter_sampling = 1000,
  refresh = 0,show_messages = F)
  ci <- quantile(fit$draws("slope"), c(0.1,0.9))
return(ci)
}
iteration = 20; intercept = -1 ; slope = 0.01
covered <- numeric(iteration)
for (i in 1:iteration){
  ci <- simulate_data(200,intercept, slope)
  covered[i] <- (slope >= ci[1] & slope <= ci[2])
}
mean(covered)
```

```
## [1] 0.85
```